Tool Catalogue and JSON Schemas SME Procurement-Preparation Agent Week 4

Tool Catalogue

Tool name	Purpose	Side effect	authorisation	Risk level
get_inventory	Retrieve current stock and reorder information for a given item	No	Any authenticated user	Low
get_quotations	Retrieve valid supplier quotations for a given item	No	Any authenticated user	Low
create_draft_requisition	Create a new purchase requisition in “Draft – Awaiting Approval” status	Yes	Allowed	Low to medium

Detailed Tool Specifications and JSON Schemas

get_inventory

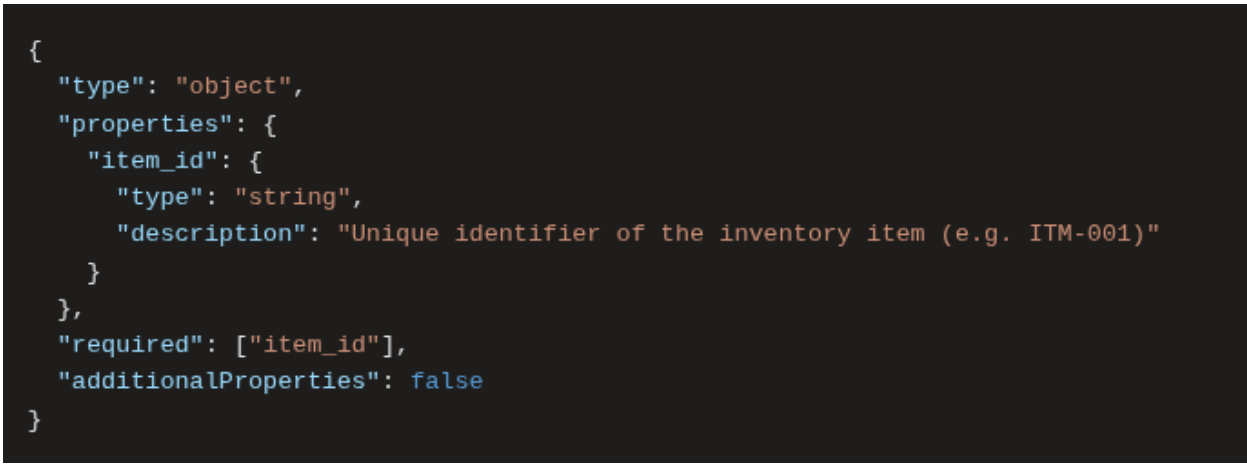
Description
Retrieves the current inventory record for a specific item, including stock level, reorder point, and recommended quantity.

Authorization
Allowed for any authenticated user. Read-only.

Failure Behaviour

- Item not found, return clear error
- Missing required parameter, return validation error
- Data source unavailable, return service error

Input Schema



```
{
  "type": "object",
  "properties": {
    "item_id": {
      "type": "string",
      "description": "Unique identifier of the inventory item (e.g. ITM-001)"
    }
  },
  "required": ["item_id"],
  "additionalProperties": false
}
```

Output schema



```
{
  "type": "object",
  "properties": {
    "item_id": {"type": "string"},
    "item_name": {"type": "string"},
    "current_stock": {"type": "number"},
    "reorder_point": {"type": "number"},
    "reorder_qty": {"type": "number"},
    "avg_monthly_usage": {"type": "number"},
    "lead_time_days": {"type": "number"},
    "unit": {"type": "string"},
    "status": {"type": "string", "enum": ["OK", "REORDER_REQUIRED"]}
  },
  "required": ["item_id", "item_name", "current_stock", "reorder_point", "reorder_qty
}
```

get_quotations

Description

Retrieves all currently valid supplier quotations for a specific item, ranked by unit price.

Authorization

Allowed for any authenticated user. Read-only.

Failure Behaviour

- No quotations found, return empty list with message
- Item not found, return error
- All quotations expired, return message indicating no valid quotations

Input Schema

```
{
  "type": "object",
  "properties": {
    "item_id": {
      "type": "string",
      "description": "Unique identifier of the item"
    }
  },
  "required": ["item_id"],
  "additionalProperties": false
}
```

Output schema

```
{
  "type": "object",
  "properties": {
    "item_id": {"type": "string"},
    "quotations": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "quotation_id": {"type": "string"},
          "supplier_name": {"type": "string"},
          "unit_price": {"type": "number"},
          "currency": {"type": "string"},
          "lead_time_days": {"type": "number"},
          "valid_until": {"type": "string", "format": "date"},
          "is_valid": {"type": "boolean"}
        }
      }
    },
    "message": {"type": "string"}
  },
  "required": ["item_id", "quotations"]
}
```

create_draft_requisition

Description

Creates a new purchase requisition record with status “Draft Awaiting Approval”.
This is a low-risk side effect. The draft cannot proceed to ordering without human approval.

Authorization

Allowed for authenticated procurement users.
Final approval / status change to “Approved” is not permitted through this tool.

Failure Behaviour

- Missing required fields, validation error
- Invalid item_id, error
- Attempt to set status other than Draft, rejected

Input Schema

```
{
  "type": "object",
  "properties": {
    "item_id": {
      "type": "string",
      "description": "Item to request"
    },
    "quantity": {
      "type": "number",
      "minimum": 1,
      "description": "Quantity to request (positive integer)"
    },
    "justification": {
      "type": "string",
      "description": "Reason for the requisition"
    },
    "requested_by": {
      "type": "string",
      "description": "Name or ID of the person creating the draft"
    }
  },
  "required": ["item_id", "quantity", "justification", "requested_by"],
  "additionalProperties": false
}
```

Output schema

```
{
  "type": "object",
  "properties": {
    "requisition_id": {"type": "string"},
    "item_id": {"type": "string"},
    "quantity": {"type": "number"},
    "status": {"type": "string", "enum": ["Draft – Awaiting Approval"]},
    "created_at": {"type": "string", "format": "date-time"},
    "message": {"type": "string"}
  },
  "required": ["requisition_id", "item_id", "quantity", "status", "created_at"]
}
```

Authorization and Safety Rules

- get_inventory and get_quotations are Read-only, freely callable.
- create_draft_requisition are Allowed but the created record is locked in Draft status.
- Any attempt to:
    - Change status to “Approved”
    - Place an order
    - Select a final supplier

      Must be blocked either by not exposing such a tool or by hard authorization checks.